MIL-D-28963A(MC)
4 December 1975
SUPERSEDING
MIL-D-28963(MC)
28 December 1970

MILITARY SPECIFICATION

DIPLEXER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers requirements for one type diplexer to provide duplex operation of an antenna in the 30 to 76 megahertz (MHz) band.

2. APPLICABLE DOCUMENTS

*2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-T-152	- Treatment, Moisture And Fungus-Resistant, Of Communications, Electronic And Associated Electrical Equipment.
MIL-T-704	- Treatment And Painting Of Materiel.
MIL-P-15024	- Plate, Tags and Bands For Identification Of Equipment.
MIL-E-16400	- Electronic, Interior Communication And Navigation Equipment, Naval Ship and Shore, General Specification For.
MIL-E-17555	- Electronic And Electrical Equipment, Accessories, And Repair Parts, Packaging And Packing Of.
MIL-E-21981	- Electronics Type Designations, Identification Plates And Markings, Requirements For.

STANDARDS

MILITARY

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T21:08Z
Check the source to verify that this is the current version before use.

Table I - Tests

Test	Reqt Para	Test Paragraph	First Article	Production	Sample	Test Number
Vibration	3.6.2	4.7.1	X		X	1
Shock	3.6.2	4.7.2	X			2
Drop	3.6.2	4.7.3	X			3
Immersion	3.6.2	4.7.4	X		X	4
High Temperature	3.6.2	4.7.5	X		X	5
Low Temperature	3.6.2	4.7.6	X		X	6
Temperature Shock	3.6.2	4.7.7	X			7
Humidity	3.6.2	4.7.8	X			8
Low Pressure		4.7.9	X			9
Salt-Fog	3.6.2	4.7.10	X			10
Sand and Dust	3.6.2	4.7.11	X			11
Inherent Isolation	3.3.1.3.1	4.7.12	X	X	X	12
Operational Isolation	3.3.1.3.2	4.7.13	X	X	X	13
Insertion loss	3.3.1.4	4.7.14	X	X	X	14
Voltage Standing Wave Ratio	3.3.1.5	4.7.15	X	X	X	15
Power Handling	3.3.1.2	4.7.16	X		X	16
Fungus	3.2.2	4.7.17	X		X	17
Reliability	3.4	4.7.18	X			18

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T21:08Z
Check the source to verify that this is the current version before use.

accordance with MIL-STD-105, inspection level S-1, acceptable quality level 6.5, and utilizing single sampling plan. The formation of the lots or batches and the lot or batch size shall consist of the following units of product.

Lot No.	Units of Product	Lot Size
1	1-15	15
2	16-45	30
3	46-100	55
4	101-200	100
5	201-400	200
6	401-1000	600
7	1001-2000	1000

One lot number for each 1000 units thereafter

4.6 Acceptance. With exception of the first article(s), acceptance of all diplexers shall be by the cognizant Government inspector.

4.7 Tests. The detailed procedures for conducting tests shall be prepared by the contractor and submitted to the contracting officer for approval prior to the initiation of any official testing. Inspection and tests shall consist of all tests required to demonstrate that the diplexers submitted for approval meet all requirements of this specification. The contractor shall prepare and submit to the procuring activity copies of all test reports, after completion of tests. Test reports shall be in accordance with MIL-STD-831.

4.7.1 Vibration test. The diplexer shall be subjected to the following vibration test while in operation. The diplexer shall be affixed to the vibration table and subjected to vibration in the frequency range of 10 to 55 hertz (Hz) at an acceleration of 5 g's or 0.060 inch total excursion, whichever is the limiting value. The frequency change shall be at a rate of less than 1 Hz. The duration of the test shall be 1/2 hour in each of three planes. During this test, there shall be no rigid body resonances and upon completion of this test, there shall be no signs of internal or external physical deterioration.

4.7.2 Shock test. The diplexer shall be subjected to the following shock test while in operation. The diplexer shall be mounted rigidly on the adapter plate of a LAB package testing machine. The machine shafts shall be operated in phase and adjusted in speed to produce average peak accelerations of 10 g's with occasional peaks to 20 g's as measured on the case of the diplexer. Malfunction is permissible provided that normal operation can be restored by adjustment of front panel controls.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T21:08Z
Check the source to verify that this is the current version before use.

4.7.3 Drop test. The diplexer shall be subjected to the following drop test while nonoperating. The diplexer shall be subjected to a vertical drop onto a hard level concrete floor from a height of 30 inches, on any edge or face of the diplexer. Upon completion of this test, normal operation shall be established by tuning of the front panel controls.

4.7.4 Immersion (leakage) test. The diplexer shall be submerged in water to a depth of 3 feet measured from its highest point and held in this position for 1 hour. During the test, all panel controls shall be operated three times through their full travel. There shall be no entry of water into the diplexer and there shall be no signs of internal or external physical deterioration.

4.7.5 High-temperature test. The high-temperature test shall be in accordance with procedure I, method 501, of MIL-STD-810. The diplexer shall be operated at a temperature of not less than 60°C nor greater than 65°C.

4.7.6 Low temperature test. The low temperature test shall be in accordance with procedure I, method 502, of MIL-STD-810. The unit shall be operated at a temperature not greater than -40°C.

4.7.7 Temperature shock test. The temperature shock test shall be in accordance with procedure I, method 503, of MIL-STD-810, with exception that the maximum temperature shall be 71°C.

4.7.8 Humidity test. The humidity test shall be in accordance with procedure I, method 507, of MIL-STD-810.

4.7.9 Low pressure test. The low pressure test shall be in accordance with procedure I, method 500, of MIL-STD-810.

4.7.10 Salt-fog test. The salt-fog test shall be in accordance with procedure I, method 509, of MIL-STD-810. Salt deposits shall be removed by rinsing with tap water.

4.7.11 Sand and dust test. The sand and dust test shall be in accordance with procedure I, method 510, of MIL-STD-810.

*4.7.12 Inherent isolation. A test shall be performed to prove that the diplexer conforms to the inherent isolation requirements of 3.3.1.3.1.

*4.7.13 Operational isolation. A test shall be performed to prove that the diplexer conforms to the operational isolation requirements of 3.3.1.3.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T21:08Z
Check the source to verify that this is the current version before use.

4.7.14 Insertion loss. A test shall be performed to prove that the diplexer conforms to the insertion loss requirements of 3.3.1.4.

4.7.15 Voltage standing wave ratio. A test shall be performed to prove that the diplexer conforms to the vswr requirements of 3.3.1.5.

4.7.16 Power handling. A test shall be performed to prove that the diplexer conforms to the power handling requirements of 3.3.1.2.

*4.7.17 Fungus. The diplexer shall be tested for fungusproofness in accordance with method 508, procedure I, of MIL-STD-810.

*4.7.18 Reliability. Reliability tests to demonstrate a specified MTBF of 6,000 hours shall be performed in accordance with MIL-STD-781, test level E, test plan V, except that in all tests temperature range shall be limited to -40°C to +65°C and vibration shall be limited to a maximum of 2.2 g's ±10 percent or 0.060 inches whichever is greater, measured at the mounting points at any nonresonant frequency between 10 and 55Hz. Each diplexer used in the reliability test shall have a minimum of 1,000 hours reliability test time. On-off cycling shall consist of applying transmitted power in accordance with 3.3.1.2 during the heating cycle, and not applying power during cooling cycle. All controls shall be cycled at a rate of five cycles per hour for a minimum of 200 hours accumulated control cycling on each diplexer. Temperature cycling shall be made to conform to MIL-STD-781. Test procedures shall be designed to confirm that the diplexer continues to meet the requirements specified in 3.3.1.2, 3.3.1.3, 3.3.1.4, and 3.3.1.5 during all phases of the reliability testing. Electrical performance of the diplexer shall not be affected by any of the tests applied. Failures shall be analyzed in accordance with MIL-STD-781.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing. Preservation, packaging and packing shall be the level of shipment specified in the contract or order in accordance with MIL-E-17555 (see 6.2).

5.2 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. These diplexers are primarily intended to be employed as components of man-packed or vehicular-installed radios such as the AN/MRC-134, AN/MRC-135 or AN/TRC-166.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T21:08Z
Check the source to verify that this is the current version before use.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

6.2.1 Procurement requirements.

(a) Title, number and date of this specification.

(b) When first article is not required (see 3.1).

(c) When first article shall consist of other than quantity of three diplexers (see 3.1).

(d) Levels of preservation, packaging and packing (see 5.1).

6.2.2 Contract data requirements. Data conforming to Data Item Descriptions DI-E-4527, DI-E-4528, DI-E-4542, DI-R-4803, DI-T-4901, DI-T-4902, DI-T-4903, DI-T-4904 and DI-V-4961 will usually be required for delivery in connection with this specification. When so required, such data will be specified for delivery on a DD Form 1423 included in the contract.

*6.3 Color. Color chips for Marine Corps green, No. 23, may be obtained from the Commandant of the Marine Corps, (Code LMO-1), Headquarters U. S. Marine Corps, Washington, D. C. 20380.

6.4 First article. The first article shall be examined and tested for approval at the contractor's plant or at an independent commercial testing laboratory acceptable to the procuring activity.

*6.5 Marginal notations. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in this notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Preparing activity:
Navy - MC
Project No. 5985-N445

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T21:08Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

2-1

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T21:08Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID
DOD-317

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commandant of the Marine Corps
Headquarters Marine Corps
(Code LMC-1)
Washington, D. C. 20380

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T21:08Z
Check the source to verify that this is the current version before use.